Domain Justification — backend-swe-opt-triage.md

Who Uses This Mode and In What Situation

An F-1 master's student — MS Information Systems or MS Computer Science at a US university — with OPT authorization starting within the next 12 months, targeting backend software engineering roles across all company stages. The student has 1–3 years of prior internship or full-time experience in Python or Node.js backend development and is evaluating 20–50 companies simultaneously, with no reliable way to distinguish which will sponsor H-1B without spending application time to find out.

The stakes are specific. OPT provides 12 months of work authorization (extendable to 36 months for STEM). The H-1B cap lottery runs once per year with approximately a 35% selection rate. A student who joins a company that does not sponsor, or one that sponsors too slowly, loses not just the job but the visa window itself. The cost of a wrong application decision is not a rejection — it is months of OPT time spent at a company that will not carry the student through the lottery.

What Information Asymmetry This Mode Addresses

Without this mode, the backend SWE student evaluating 30 companies faces four invisible gaps that the engine's data layers can partially or fully close:

Gap 1 — Sponsorship history is not surfaced at the job posting level.
Job postings on LinkedIn, Indeed, and company career pages do not display H-1B petition history. A student cannot distinguish a company with 50 SWE LCA filings since 2022 from one with zero. This information exists in DOL LCA disclosure data but is not aggregated at the company-plus-SOC-code level anywhere publicly visible without manual research per company. The 80 Days dataset closes this gap.

Gap 2 — Startup viability is not visible from the career page.
A seed-stage company with a polished Ashby career page and a well-written job description may have Form D funding that is 30 months old and below the threshold

for meaningful runway. A student applying to 10 such companies is spending OPT clock on employers that may not exist at H-1B lottery time. The Form D signal in the 80 Days layer closes this gap for disclosed raises; it does not close it for burn rate or undisclosed down rounds.

Gap 3 — Ghost listings consume application effort before any human sees them. ATS-posted roles that have been filled or frozen are not marked expired on most job boards. The Job-Ops liveness checker closes this gap by running a Playwright check against the posting URL before the student drafts anything.

Gap 4 — Role longevity under AI substitution is not visible from the posting. An OPT student is evaluating a role not just for the next 12 months but through H-1B renewal (3–6 years out). Backend roles vary on this dimension: platform infrastructure and distributed systems roles score high on cognitive demand and system judgment; high-volume CRUD and scaffolding roles score lower and face meaningful automation pressure. The Cognitive Pivot layer closes this gap at the SOC level; within-SOC differentiation requires manual review.

Connection to Engine Layers

80 Days to Stay is the load-bearing layer. The H-1B LCA disclosure data filtered to SOC 15-1252 and adjacent codes directly answers Gap 1. The Form D recency data answers Gap 2 for pre-IPO companies. The 30K+ company dataset makes the filter meaningful at scale rather than a per-company manual lookup.

Job-Ops closes Gap 3 directly. The Playwright liveness checker and ATS detector prevent application effort from being spent on expired postings or career pages returning no active roles. The connection to `data/ats/pipeline.md` also prevents the student from triaging companies already in their active pipeline.

The Cognitive Pivot addresses Gap 4 at the SOC level. The limitation is real and stated in the mode: the score for SOC 15-1252 does not distinguish a distributed systems role from a CRUD API role. The mode flags this as inferred judgment, not verified data, and instructs the student to review role descriptions manually.

Failure Modes

Failure Mode 1 — Company has H-1B history but stopped sponsoring after layoffs. The LCA disclosure dataset reflects historical filings. A company that laid off 30% of its engineering team in 2023 and stopped filing new H-1B petitions will still show historical petition counts from before the policy change. The mode will classify it as Tier 1 based on counts that no longer reflect current intent.

This error is hardest to catch for a student who does not cross-reference LCA data with layoff databases (Layoffs.fyi) or community forums (Blind). The failure is not in the data — the data is accurate. The failure is in treating a historical filing count as a forward-looking signal without checking for a gap between historical volume and recent filings. The mode's proposed `h1b_recency.py` script would surface this gap automatically; until it exists, a declining year-over-year petition count is the signal to check manually.

Failure Mode 2 — SOC code mismatch produces a false negative for a real sponsor. Backend engineering roles are not consistently filed under SOC 15-1252. Some employers file "Platform Engineer" under 15-1244 (Network and Computer Systems Administrators) or "API Engineer" under 15-1299 (Computer Occupations, All Other). A filter on 15-1252 alone will return zero petitions for these companies even when they are active sponsors.

This is the hardest failure to catch because it produces a false negative: the mode tiers a legitimate sponsor as Tier 3 or unverified, and the student deprioritizes or skips a company that would have been a strong match. Running the filter across all three SWE-adjacent SOC codes (as the mode's Step 2 recipe now specifies) reduces but does not eliminate this risk.

Failure Mode 3 — Form D recency is treated as runway confirmation. A student applying to a startup with a Form D filing 11 months ago and a $2M raise is at real risk of the company not surviving to the H-1B lottery. The mode will tier this company as Tier 2 (funding signal present, insufficient H-1B history) — which is a reasonable triage classification but not a viability endorsement.

This error is hardest to catch for students unfamiliar with startup financial signals, who read a recent Form D as evidence of ongoing health rather than as a timestamp of a single capital event. The mode explicitly flags this in the Cannot Verify section; it does not solve it, because burn rate and current runway are not in the dataset.